BOYANG XIAO

Mechanical / Robotics Engineer | Surgical Robotics | Medical Device R&D

boyang.xiao@vanderbilt.edu | +1-410-814-1773 | www.linkedin.com/in/boyang-xiao

PROFESSIONAL SUMMARY

Ph.D. candidate in Mechanical Engineering with 5+ years of experience designing, modeling, integrating, and experimentally validating medical robotic systems for minimally invasive applications. Experienced in translating clinical needs into device requirements and test protocols, developing robotic actuation and sensing platforms, and implementing calibration and control workflows. Collaborated with clinicians on anatomical phantom and ex vivo testing in gastrointestinal and urological applications, with additional experience in precision robotics for retinal surgery.

CORE TECHNICAL SKILLS

Robotics & Controls	kinematics, trajectory tracking, position control, kinematic calibration, soft robots, magnetic actuation, Cosserat-Rod modeling, robotic-arm control, Robot Operating System (ROS 1/2), Franka robotic arm
Integration & Testing	system integration, AprilTag tracking, color-based image segmentation, test protocol development, benchtop test fixtures, anatomical phantom and ex vivo testing, statistical analysis
Software & Test	C++, MATLAB, Python, ROS, LabVIEW, Arduino, SolidWorks; miniature robotic mechanisms, sensors, motor drivers, 3D printing, laser micromachining

EDUCATION

Vanderbilt University — Nashville, TN
Ph.D. in Mechanical Engineering | Advisor: Prof. Xiaoguang Dong — Jul 2022 – expected Jan 2027

Johns Hopkins University — Baltimore, MD
M.S. in Mechanical Engineering | Advisor: Prof. Iulian Iordachita — Jan 2020 - Dec 2021

Shanghai Jiao Tong University — Shanghai, China
B.S. in Engineering Mechanics — Sep 2015 - Jun 2019

PATENT

- Dong, X. and **Xiao, B.** (2025). Viscosity sensors and wireless small-scale soft robots. U.S. Patent Application No. 18/882,837.

RESEARCH AND DEVELOPMENT EXPERIENCE

Research Assistant — Jul 2022 - Present
Dong Lab, Department of Mechanical Engineering
Vanderbilt Institute for Surgery and Engineering (VISE), Vanderbilt University

Project 1: Soft Robotic Catheter Control and Navigation for Minimally Invasive Interventions

- Designed and fabricated a wheel-based soft magnetic robotic catheter for navigation through narrow, branching anatomical pathways, with potential applications in biliary and endovascular interventions.
- Demonstrated wireless magnetic navigation in biliary ducts as narrow as 2 mm, negotiating turns up to 150° and selectively entering branches at anatomical bifurcations.
- Derived kinematic and dynamic models using Cosserat rod theory to predict catheter deformation and locomotion under magnetic actuation, supporting device design and navigation control.

- Developed a Franka robotic arm-based magnetic actuation platform integrating magnetic field prediction, robot tracking, and position control for repeatable navigation experiments.
- Developed AprilTag- and color-segmentation-based tracking of a magnetic actuation system and soft robot for endoluminal navigation, supporting preparation of the Open-H-Endoluminal dataset proposal.
- Designed and fabricated a custom delivery catheter compatible with endoscopic retrograde cholangiopancreatography (ERCP) procedures; validated robot delivery and retrieval in an anatomical phantom.
- Collaborated with a gastroenterologist to validate navigation in a 3D biliary-tract phantom and ex vivo porcine anatomy, evaluating performance in clinically relevant anatomical environments.

Project 2: Wireless Microfluidics-Integrated Miniature Soft Robots for Gastrointestinal Applications

- Designed and fabricated magnetic miniature soft robots with multimodal locomotion, including climbing, crawling, and rolling to access hard-to-reach anatomical regions with conventional endoscopes.
- Developed a wireless magnetic soft sensor for spatiotemporal in situ mucus-viscosity sensing as a potential gastrointestinal diagnostic tool.
- Integrated magnetically actuated wireless pumping and valving mechanisms to enable liquid release, biopsy, and sampling functions for gastrointestinal and confined-duct applications.
- Modeled the soft magnetic robot deformation under magnetic fields using Cosserat Rod theory and beam theory to predict and control robotic behavior.

Project 3: Wireless Implantable Device with Flexible Magnetic Valves for Bladder Disease Treatment

- Developed and demonstrated the simultaneous control of multiple magnetic valves integrated into a soft robotic patch for intravesical drug delivery.
- Integrated a resonant-frequency-based sensing method using a vector network analyzer to estimate released liquid volume and enable closed-loop regulation of on-demand drug release.
- Modeled soft magnetic valve behavior and magnetic field distribution to predict valve opening/closing thresholds.
- Mentored a master student on related minimally invasive valve therapy.
- Collaborated with a urologist to validate closed-loop drug release in an ex vivo porcine bladder model.

Research Assistant, AMIRo Lab, Johns Hopkins University May 2021 - Jun 2022

Precision Robot Calibration and Control for Robot-Assisted Retinal Surgery

- Calibrated Eye Robot 3.0's Delta robot and tilting mechanism with a 21-parameter kinematic model and Bernstein polynomial compensation; tool-tip error <20 micrometers.
- Developed cooperative control with a remote center of motion (RCM) constraint to support hands-on manipulation of the surgical instrument.
- Implemented inverse kinematics, Cartesian position control, and experimental data-logging functions using ROS and CISST to support repeatable robot operation and experimental validation.

REPRESENTATIVE PUBLICATIONS

Google Scholar

1. **Xiao, B.**, …& Dong, X. (2024). Sensing mucus physiological property in situ by wireless millimeter-scale soft robots. *Advanced Functional Materials,* 34(8), 2307751.
2. **Xiao, B.**, …, Tosoian, J. J. & Dong, X. (2026). Fully wireless and flexible valves for multiplexed and prolonged intravesical liquid release. *Advanced Healthcare Materials*, 15(23). **Back Cover**.
3. Xu, Y.*, **Xiao, B.*** (*co-first authors), …, Obstein, K. L & Dong, X. (2023). Wireless millimeter-size soft climbing robots with omnidirectional steerability on tissue surfaces. *IEEE Robotics and Automation Letters,* 8(9), 5720-5726.
4. **Xiao, B.**, …, Tosoian, J. J. & Dong, X. (2025). Wireless microfluidics-enabled multifunctional miniature soft robots with multimodal locomotion for fluid manipulation. *Device,* 3(6).
5. **Xiao, B.**, …, Gehlbach, P., Taylor, R. H., & Iordachita, I. (2022). Delta robot kinematic calibration for precise robot-assisted retinal surgery. *International Symposium on Medical Robotics (ISMR)*, 1-7.